

Practice Problems Week 1

1. Make a point a line graph that reflects the following raw network data

A and D talk.

A and B talk.

C and E talk.

D and C talk.

B and C talk.

D and E talk

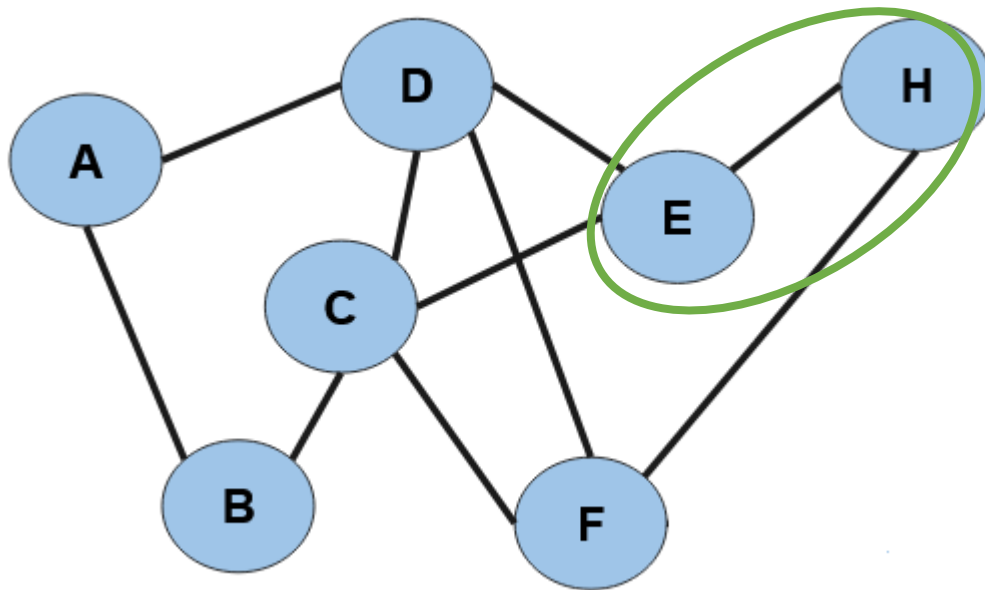
C and F talk

D and F talk

F and H talk.

E and H talk.

Added because I forgot to add it to the original problem



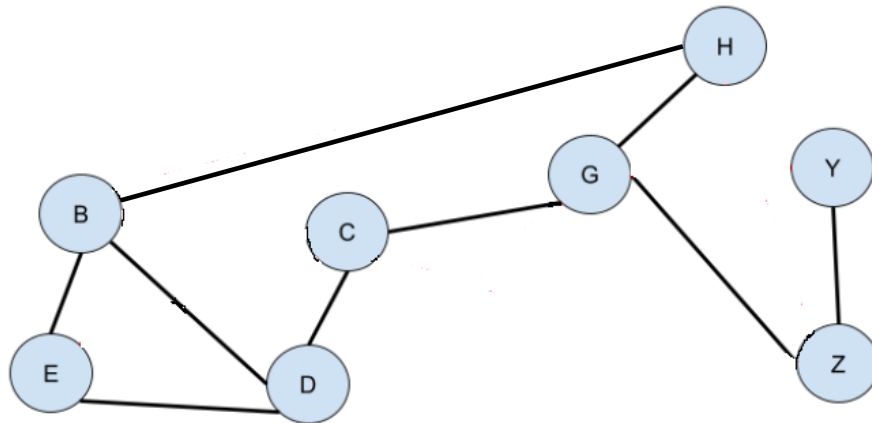
2. What is the set that corresponds to the above data?

$G = \{V, E\}$

$V = \{V_A, V_B, V_C, V_D, V_E, V_F, V_H\}$

$E = \{V_A V_D, V_A V_B, V_C V_E, V_D V_C, V_B V_C, V_D V_E, V_C V_F, V_D V_F, V_F V_H, V_E V_H\}$

3. What is the shortest path from E to G? B to Z? Y to E?



E-G={EB, BH, HG}

E-G={ED, DC, CG}

Path length 3 (extra, wasn't asked in the question)

B-Z={BH,HG,GZ}

Path length 3 (extra, wasn't asked in the question)

Y-E={YZ, ZG, GC, CD, DE}

Y-E={YZ, ZG, GH, HB, BE}

Path length 5 (extra, wasn't asked in the question)